FYP24-030-R-SurgiGuide

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Chapter 1

Introduction

This study aims to create an AI-based decision support system named SurgiGuide that addresses the ongoing issue of preventable surgical mistakes. The problem domain for this study comprises the continued occurrence of such errors even with ongoing technological improvements in surgical practices, the challenges surgeons face when making decisions in high-stakes environments, the lack of consistent practices and objective methods of assessing a surgeon's competencies, and the potential to leverage technology to improve the learning of surgical procedures.

The research question focuses on establishing a mechanism for identifying the phases of a surgical process in real-time, providing accurate phase analysis, identifying possible errors, offering feedback, and suggesting corrective actions based on commonly known best practices.

Through fulfilling these aims, SurgiGuide seeks to increase overall awareness during surgery, promote better surgical outcomes, improve surgical operations' operational efficiency, and enhance surgeons' growth, all of which ultimately contribute to better patient care.

1.1 Problem Domain

The problem statement of SurgiGuide is about surgical decision making and minimisation of errors through the use of artificial intelligence (AI). Surgeries are complicated operations that require significant mental and manual effort from the side of surgeons. They frequently entice important choices with active threats for the patient and may have clinical effects within the organization.

People's mistakes still play a vital role in defining the outcomes of operations, even with new techniques and instruments. Wrong-site surgeries and retained surgical items or sponges are examples of “Never Events,” which again evidence that the safety of the operating room environment and the decision support systems used need to be enhanced.

Stakeholders argue that surgeons need better tools and systems that would help control the surgery in real time, inform the surgeon of possible dangers, and contribute to the decision-making process.

1.2 Research Problem Statement

The potential for surgical decision-making and errors is intended to be solved by creating AI real-time surgical guidance named SurgiGuide. This system will:

- **Identify the current phase of surgery**: By identifying the various phases of a surgical process from a live video stream, SurgiGuide can produce contextually relevant data for a surgeon during the surgery. Research shows that TEP hernia repair assessment is capable of automatic recognition of surgical steps with high accuracies.
- **Suggest the next step of surgery**: Based on phase recognition, SurgiGuide will use artificial intelligence to predict the sequence of actions that should be taken next, supporting the surgical process and guiding surgeons to the right decision.
- **Point out mistakes made during the surgery or procedure**: Real-time comparison of actual surgical activities and instruments used against a database of required standards and a list of surgical mishaps will be the prime goals of SurgiGuide. This aligns with the emerging domain of surgical-data science, which seeks to identify how exactly AI can detect when a procedure deviates from the norm and warn surgeons.
- **Provide a remedy to the mistake**: If there is any indication of an error, SurgiGuide will give the surgeon recommendations for appropriate corrective adjustments to minimize the likelihood and extent of adverse events.

By integrating these functionalities, SurgiGuide seeks to:

- Improve the overall risk during surgery and the rate of avoidable mistakes.
- Optimize timing and throughput within the operating theatre environment and provide support to surgeons to enable efficient decision-making.
- Serve as an educational tool for training surgeons by providing real-time feedback.

Addressing this research problem requires overcoming several challenges, including:

- Creating highly dependable models with proven success in recognizing phases, actions, and instruments in different surgeries and environments.
- Implementing detailed databases to track surgical mistakes and develop superior techniques for the system's mistake identifier and preventer.
- Maintaining a natural user interface that integrates smoothly into the workflow of surgeons and nurses.
- Validating SurgiGuide's actual performance in live and complex surgical scenarios.

In addressing these issues, SurgiGuide seeks to make a considerable impact in the area of AI-aided surgery, thus contributing to enhancing the quality of patient care.

Chapter 2

Literature Review

This chapter critically examines existing literature relevant to the research topic. It identifies key studies, theories, and findings, providing a foundation for the proposed research and highlighting its relationship to prior work.

2.1 Related Research

- Surgical Phase Recognition in Laparoscopic Cholecystectomy[4]
- A Vision Transformer for Decoding Surgeon Activity from Surgical Videos [3]
- Factors Contributing to Preventing Operating Room 'Never Events': A Machine Learning Analysis[1]
- Automatic Surgical Skill Assessment System Based on Concordance of Standardized Surgical Field Development Using Artificial Intelligence[2]
- A Novel High Accuracy Model for Automatic Surgical Workflow Recognition Using Artificial Intelligence in Laparoscopic Totally Extraperitoneal Inguinal Hernia Repair (TEP)[5]
- Comparative Validation of Machine Learning Algorithms for Surgical Workflow and Skill Analysis with the HeiChole Benchmark[6]

2.1.1 Surgical Phase Recognition in Laparoscopic Cholecystectomy

2.1.1.1 Summary of the research item

This research paper presents a new approach for improving surgical phase identification in videos of laparoscopic cholecystectomy (LC). The authors address the limitations of existing computer vision methods in analyzing sequential processes and propose a dynamic switching mechanism between two Transformer-based models: a baseline model and a binary classifier.

The baseline model, Trans-SVNet, which has superior performance in surgical phase recognition, analyzes the entire surgical video and identifies various phases based on visual features. The binary classifier, on the other hand, focuses on unique phase transitions in the LC procedure. The system relies on calibrated confidence scores produced by the baseline model. When the confidence of the baseline model is high, it is retained as the primary model. However, when confidence drops below a certain threshold, the system switches to the binary classifier, which is specifically trained for phase transitions to provide a better estimate during those segments.

Although the paper does not provide experimental results, the authors suggest that this dynamic switching mechanism can enhance surgical phase identification and can be easily integrated with existing phase recognition techniques to improve surgical workflow analysis.

2.1.1.2 Critical analysis of the research item

Strengths:

- **Addresses Sequential Nature of Surgery:** The paper acknowledges the sequential nature of surgical procedures and leverages this characteristic to improve prediction accuracy, rather than analyzing each frame independently.
- **Dynamic Model Switching:** The dynamic switching between models based on calibrated confidence scores utilizes the strengths of each model, offering a flexible and potentially more accurate approach to surgical phase recognition.
- **Applicability to Existing Methods:** The dynamic switching mechanism proposed by the authors can be applied to any existing phase recognition model, making it versatile and potentially impactful across the field.

Limitations:

- **Lack of Experimental Results:** The paper introduces a theoretical approach without experimental validation, making it difficult to assess the practical improvements or limitations of the method.
- **Limited Information on Confidence Calibration:** Although the paper discusses the use of calibrated confidence scores, it does not provide details on how the calibration process was conducted, which is crucial for evaluating the model-switching mechanism's reliability.

- **Focus on a Single Procedure:** The paper focuses on LC, and it remains uncertain whether this approach can be generalized to other surgical procedures with varying complexities and structures.

2.1.1.3 Relationship to the proposed research work

This research aligns closely with the objectives of the SurgiGuide project, particularly in the areas of phase categorization and task direction.

- **Incorporating sequential modeling techniques:** The use of Transformers and dynamic switching based on confidence scores could improve SurgiGuide's ability to recognize surgical phases more effectively than existing methods like HeiChole.
- **Predicting next surgical steps:** Accurate phase identification is a key prerequisite for determining subsequent actions in surgery. The enhanced accuracy offered by this approach could help SurgiGuide generate more precise and timely recommendations for surgeons during operations.
- **Generalizing to diverse procedures:** While the approach shows promise in cholecystectomy, validating it on datasets from other surgical specialties could enhance SurgiGuide's general usability across diverse procedures.
- **Addressing real-time performance:** Further research into the time complexity of this algorithm and improvements for real-time application are crucial to integrating this method into SurgiGuide for live surgical support.

2.1.2 A Vision Transformer for Decoding Surgeon Activity from Surgical Videos

2.1.2.1 Summary of the research item

This research article introduces a new AI system named SAIS, which can facilitate the analysis of tasks incorporated in surgical videos and decode distinct aspects of surgical movements. SAIS can tell when a particular subphase of the procedure is performed, categorize the motions by which a task is conducted, and determine how proficiently an operation is done.

The researchers applied SAIS to videos of robot-assisted radical prostatectomy (RARP) performed by other hospitals, showing that the model performs well across various surgeons, procedures, and locations. SAIS offered high accuracy in recognizing sub-phases, gestures, and differences in surgical techniques.

Moreover, SAIS recognized the ability to distinguish between high and low skill actions, correlating well with expert opinion. It also provides explanations for skill ratings by identifying frames in the video.

2.1.2.2 Critical analysis of the research item

Strengths:

- **Unified architecture:** SAIS uses a consistent architecture to parse multiple elements of surgical activity, simplifying future modifications.
- **Explainability:** The system provides reasons for skill ratings, increasing confidence in its assessments.
- **Generalisability:** SAIS performs well across multiple hospitals and surgeons, supporting its scalability and potential for practical use.
- **Novel Insights:** SAIS creates time-series profiles of surgeons' skills, enabling personalized performance reviews and deficiency identification.

Limitations:

- **Focus on RARP:** Primarily tested on RARP videos, the effectiveness of SAIS for other procedures is unknown.
- **Limited clinical validation:** Larger studies are needed to examine the relationship between SAIS use and patient outcomes.

2.1.2.3 Relationship to the proposed research work

The research on SAIS directly aligns with several key goals of the SurgiGuide project:

- **Real-time surgical phase recognition:** SAIS's success in identifying subphases in real-time demonstrates that such functionality can be integrated into SurgiGuide.
- **Surgical error identification and correction:** SAIS's ability to identify and explain low-skill actions could inform real-time error detection and correction in SurgiGuide.
- **Data-driven insights for surgical training:** SAIS's generation of skill profiles offers an important tool for surgical training, allowing trainees to track progress and identify areas for improvement.

This research provides evidence that AI can be integrated into SurgiGuide, informing its design and implementation for safer and more efficient surgeries.

2.1.3 Factors Contributing to Preventing Operating Room 'Never Events': A Machine Learning Analysis

2.1.3.1 Summary of the research item

This research article aims to identify and calculate risk factors contributing to surgical "Never Events," which are preventable incidents occurring before, during, or after surgery. The authors used machine learning, specifically random forests, to analyze over 9,000 safety observations and 101 Never Event root-cause analyses related to wrong-site surgery and retained foreign objects.

The data came from 29 Israeli hospitals between 2016 and 2020, based on observations by trained medical practitioners and students. A comparison of these observations to root-cause analyses of Never Events led to the identification of 24 significant causal factors, including surgery duration, team members, zip code, region, and patient characteristics such as BMI and ASA score.

2.1.3.2 Critical analysis of the research item

Strengths:

- **Large, real-world dataset:** The large dataset, containing observations and root-cause analyses, increases the generalization and practical use of the research.
- **Focus on quantifiable risk:** The study identifies and quantifies the influence of risk factors, offering actionable insights for risk mitigation.
- **Machine learning approach:** The use of random forests enables understanding of the relationships between variables and predicts the likelihood of Never Events.

Limitations:

- **Limited scope of Never Events:** The research focuses on only two types of Never Events, which limits its applicability to other preventable surgical errors.
- **Geographical context:** The study is based on Israeli hospitals, so its findings may not be directly applicable to other healthcare systems.
- **Limited focus on prevention strategies:** The research identifies causes but provides few concrete recommendations for preventing Never Events.

2.1.3.3 Relationship to the proposed research work

This article's findings hold significant relevance to the SurgiGuide project, particularly in the following areas:

- **Error identification and risk mitigation:** The study complements SurgiGuide's goal of identifying risks and preventing surgical errors by outlining potential risk factors for Never Events.
- **Data-driven insights for surgical training:** The risks identified in this study can be used to enhance SurgiGuide's training modules, helping surgical teams improve decision-making.
- **Real-time feedback and decision support:** Incorporating these risk factors into SurgiGuide could enable real-time identification and correction of potential Never Events during surgery.

By applying the study's insights to SurgiGuide, the system could help reduce surgical risks and improve safety in the operating room.

2.1.4 Comparative Validation of Machine Learning Algorithms for Surgical Workflow and Skill Analysis with the HeiChole Benchmark

2.1.4.1 Summary of the research item

This research paper introduces the HeiChole benchmark, a new public dataset designed for assessing surgical performance and tasks in laparoscopic cholecystectomy (LC). The study uses this dataset to evaluate and compare the performance of various machine learning (ML) algorithms in automatically recognizing key surgical elements from LC videos, specifically:

- **Surgical Phases:** Subprocesses in the surgery (e.g., preoperative preparation, surgery).
- **Surgical Actions:** Specific actions performed by the surgeon during the procedure, such as grabbing, incising, clasping, or pinching.
- **Surgical Instruments:** Identification of surgical instruments like graspers or clippers used during the procedure.
- **Surgical Skill:** Evaluation of the surgeon's expertise based on video material.

The HeiChole dataset consists of 33 LC videos, collected from three surgical centers, amounting to 22 hours of operating time. These videos are meticulously annotated, frame by frame, for the four surgical aspects mentioned above, covering 250 phase transitions, 5514 actions, 6980 instruments, and 495 skills. The annotation process involved multiple medical students following a well-defined protocol to minimize variability.

This work provided the foundation for the EndoVis Challenge in 2019, with a sub-challenge focused on surgical workflow and skill assessment. Ten teams participated, applying their ML algorithms to the HeiChole dataset. The challenge evaluated the algorithms' performance in identifying surgical phases, actions, tools, and skill levels.

Key Findings:

- Performance varied significantly across ML algorithms and recognition tasks.
- Phase recognition achieved an accuracy of 23.9-67.7% (F1-score), while instrument recognition ranged from 38.5-63.8% (F1-score).
- Action recognition proved more challenging, with F1-scores between 21.8 and 23.3.
- Only one team completed the skill assessment task, with an average absolute error of 0.78.

2.1.4.2 Critical analysis of the research item

Strengths:

- **Open Dataset and Benchmark:** The HeiChole benchmark, with highly detailed annotations, is a significant contribution to surgical data science. Its public availability allows for algorithm comparisons and helps enhance surgical workflow evaluation using ML.
- **Multi-Centre Data Collection:** The dataset's collection from three centers introduces diversity and captures variations in surgical practices and patient populations.
- **Focus on Challenging Tasks:** By assessing surgical actions and skill, in addition to phases, the study addresses deeper issues in surgical workflow that go beyond simple phase recognition, pushing the limits of current ML approaches.
- **Detailed Annotation Protocol:** The well-structured annotation process, involving multiple coders, ensures high-quality, reliable labels, which is crucial for the dataset's validity.

Limitations:

- **Limited Number of Participants:** Few teams participated, particularly in the action and skill assessment tasks, reducing the statistical power and generalizability of the findings.

- **Specificity of the Dataset:** Despite LC being a common procedure, the results may not generalize to more complex surgeries. Creating benchmarks for other procedures is vital for broader applicability.

- **Reliance on Visual Data Alone:** The study focuses solely on video analysis, ignoring other valuable data sources like surgical instrument kinematics or physiological signals, which could enhance the analysis.

- **Subjectivity in Skill Assessment:** Assessing surgical skill involves subjective judgment, even among human experts. The paper acknowledges the limitations of using a modified GOALS score and the potential bias in annotator ratings.

2.1.4.3 Relationship to the proposed research work

The findings of this research directly relate to the objectives of SurgiGuide, which aims to use AI to improve surgical performance by:

- **Pinpointing surgical phases:** The study's findings on phase recognition (up to 67.7% F1-score) provide a foundation for SurgiGuide to accurately identify phases in real-time, enhancing surgical workflow understanding.

- **Suggesting next surgical steps:** Accurate phase identification is a prerequisite for sequence anticipation, which will help SurgiGuide guide surgeons and potentially reduce errors during procedures.

- **Identifying mistakes:** Although current ML models struggle with action and skill recognition, the HeiChole benchmark provides a resource for improving these capabilities in SurgiGuide, enabling the system to detect surgical mistakes.

- **Providing remedies for mistakes:** As action and skill recognition models improve, SurgiGuide could suggest corrective actions to surgeons in real time, improving patient safety.

2.1.5 Automatic Surgical Skill Assessment System Based on Concordance of Standardized Surgical Field Development Using Artificial Intelligence

2.1.5.1 Summary of the research item

This research paper deals with the formulation and performance analysis of a deep learning model for laparoscopic sigmoid colon resection (Lap-S) surgical skill assessment. The model centers on identifying the uniformity in creating the surgical field over the duration of the procedure by calculating a new parameter called the AI Confidence Score (AICS).

The researchers used 650 intraoperative videos of Lap-S that were submitted to the JSES for the ESSQS between August 2016 and November 2017. ESSQS is an established system in Japan that assesses the surgical competence of endoscopic surgeons.

For model training, 60 videos of surgeons with an ESSQS score greater than 75 were chosen to ensure that the model was trained using standardized surgeries. The model assesses how the surgical field is built in each phase of the Lap-S operation and compares this with the standardized approach. Based on this comparison, the model calculates an AICS for each video, representing the degree of correspondence between the development of the surgical field and the expert surgeons' gold standard. A higher AICS indicates greater alignment with the standardized surgical field development.

The research revealed a positive correlation between AICS and the ESSQS scores assigned to the videos (Spearman rank correlation coefficient = 0.81, $P < .001$). This finding supports the idea that AICS can serve as an objective and automated substitute for human assessment of surgical skill in Lap-S. Furthermore, the model demonstrated high accuracy in offline automatic screening for low- and high-score ESSQS groups, solidifying its potential for use in robotic surgical skill assessment models.

2.1.5.2 Critical analysis of the research item

Strengths:

- **Objective Skill Assessment:** The AI-based model offers a more objective assessment compared to human experts, who may introduce substantial inter-subjective variations.

- **Focus on Surgical Field Development:** The study emphasizes standardized surgical field development as an important aspect of skill assessment, alongside the instruments and tools used in the procedure.

- **Strong Correlation with Established System:** The strong correlation between AICS and the widely-used ESSQS scores suggests the model's validity and potential clinical relevance.
- **Potential for Automated Screening:** The model's high specificity and sensitivity in differentiating between low and high ESSQS score groups indicate its potential for automating the initial screening of surgical skills.

Limitations:

- **Limited Scope:** The study focuses solely on Lap-S, and the model's generalizability to other laparoscopic procedures or surgical specialties remains to be determined.
- **Single-Institution Data:** The study used videos from only one institution (JSES), which may introduce bias and limit the model's applicability in other regions.
- **Limited Sample Size:** Although 650 videos were collected, only 60 were used for training and 60 for validation. Expanding the dataset with more diverse sources is needed to improve the model's stability and usability.
- **Focus on Visual Cues Only:** The model only analyzes the surgical field, without incorporating data such as surgical instrument kinematics or physiological signals, which could improve skill assessment.
- **Lack of Feedback Mechanism:** While the model measures surgical skill, it does not provide feedback on specific areas of concern, which is essential for surgeon training.

2.1.5.3 Relationship to the proposed research work

This research aligns with several key objectives of the SurgiGuide project:

- **Provide feedback on surgical performance:** The processes used by the AI system to evaluate surgical skills match SurgiGuide's objective to offer real-time feedback to surgeons during surgery.
- **Identify mistakes and suggest corrections:** Just as SurgiGuide aims to identify surgical mistakes, this research highlights AI's ability to recognize deviations from standard practices, providing a foundation for suggesting corrective actions.
- **Assist in surgical training:** The AI system's capability for screening and evaluating surgical skills may prove useful in SurgiGuide's mission to enhance surgical education and training by gathering more information on trainee performance.

2.1.6 A Novel High Accuracy Model for Automatic Surgical Workflow Recognition Using Artificial Intelligence in Laparoscopic Totally Extraperitoneal Inguinal Hernia Repair (TEP)

2.1.6.1 Summary of the research item

This study focuses on the development and application of an AI model aimed at recognizing surgical workflows during laparoscopic totally extraperitoneal inguinal hernia repair (TEP). The research evaluated the performance of the model using 619 TEP procedure videos, with 371 for training, 93 for internal validation, and 155 for testing.

The AI model demonstrated an overall accuracy of 88.8% in recognizing the six critical steps of TEP, with individual step accuracies ranging from 72.2% to 94.3%. The model performed best in recognizing the hernia sac reduction step and faced more challenges during the preperitoneal dissection step.

2.1.6.2 Critical analysis of the research item

Strengths:

- **High accuracy:** The model's high accuracy in identifying different stages of TEP indicates the strong potential of AI in automating surgical workflow recognition.
- **Specificity to TEP repair:** The model's focus on a specific procedure enhances the precision of step recognition, making it more reliable for TEP surgeries.
- **Large and diverse dataset:** The use of 619 videos from multiple surgeons with different techniques contributes to the model's robustness and versatility.

Limitations:

- **Limited generalizability:** The dataset, though diverse, is primarily drawn from a single medical center, which limits the model's applicability to other types of surgeries or different clinical environments.
- **Simplified step definition:** The study uses standardized surgical steps, but these may oversimplify the variability present in real-world procedures, potentially reducing the model's effectiveness in more complex cases.
- **Lack of external validation:** The absence of external validation raises concerns about the model's performance in varied clinical settings.

2.1.6.3 Relationship to the proposed research work

This study provides direct insights into how AI can enhance the objectives of the SurgiGuide project:

- **Real-time step recognition:** The AI model's ability to recognize the stages of TEP procedures with high accuracy could be applied to SurgiGuide's aim of offering real-time surgical phase identification.
- **Surgical error reduction:** The step-by-step recognition model could help SurgiGuide track deviations from standard procedures, enabling the system to identify potential errors and offer corrective feedback.
- **Data-driven insights and training:** The dataset and AI's step recognition could be utilized within SurgiGuide to provide automated feedback and training for surgeons, helping improve performance and reduce mistakes.

Incorporating these findings into SurgiGuide's framework could strengthen its capacity to assist surgeons during TEP procedures and beyond, leading to improved surgical safety and efficiency. Comparative Validation of Machine Learning Algorithms for Surgical Workflow and Skill Analysis with the HeiChole Benchmark

2.2 Analysis Summary of Research Items

Table 2.1: Analysis of Related Research

Research Item	**Key Findings**	**Strengths**	**Limitations**
Surgical Phase Recognition in Laparoscopic Cholecystectomy	Dynamic switching mechanism between Transformer-based models enhances surgical phase identification.	• Addresses the sequential nature of surgery. • Dynamic model switching. • Applicability to existing methods.	• Lack of experimental results. • Limited information on confidence calibration. • Focus on a single procedure.

Research Item	Key Findings	Strengths	Limitations
A Vision Transformer for Decoding Surgeon Activity from Surgical Videos	SAIS effectively analyzes surgical movements and identifies skill levels across various procedures.	• Unified architecture. • Explainability. • Generalizability. • Novel insights into surgical performance.	• Primarily tested on RARP. • Limited clinical validation.
Factors Contributing to Preventing Operating Room 'Never Events': A Machine Learning Analysis	Identifies risk factors contributing to preventable surgical incidents using a large dataset.	• Large dataset. • Focus on quantifiable risk. • Machine learning approach enhances understanding.	• Limited scope of Never Events. • Geographical context. • Limited focus on prevention strategies.
Comparative Validation of Machine Learning Algorithms for Surgical Workflow and Skill Analysis with the HeiChole Benchmark	HeiChole dataset provides a benchmark for recognizing surgical elements from laparoscopic videos.	• Open dataset. • Multi-centre data collection. • Focus on challenging tasks. • Detailed annotation protocol.	• Limited number of participants. • Specificity of the dataset. • Reliance on visual data alone. • Subjectivity in skill assessment.

Research Item	Key Findings	Strengths	Limitations
Automatic Surgical Skill Assessment System Based on Concordance of Standardized Surgical Field Development Using Artificial Intelligence	AI Confidence Score correlates with established surgical competence metrics.	• Objective skill assessment. • Focus on surgical field development. • Potential for robotic skill assessment.	• Requires further validation across diverse surgical contexts. • Reliance on historical data may limit applicability.

Chapter 3

Proposed Approach

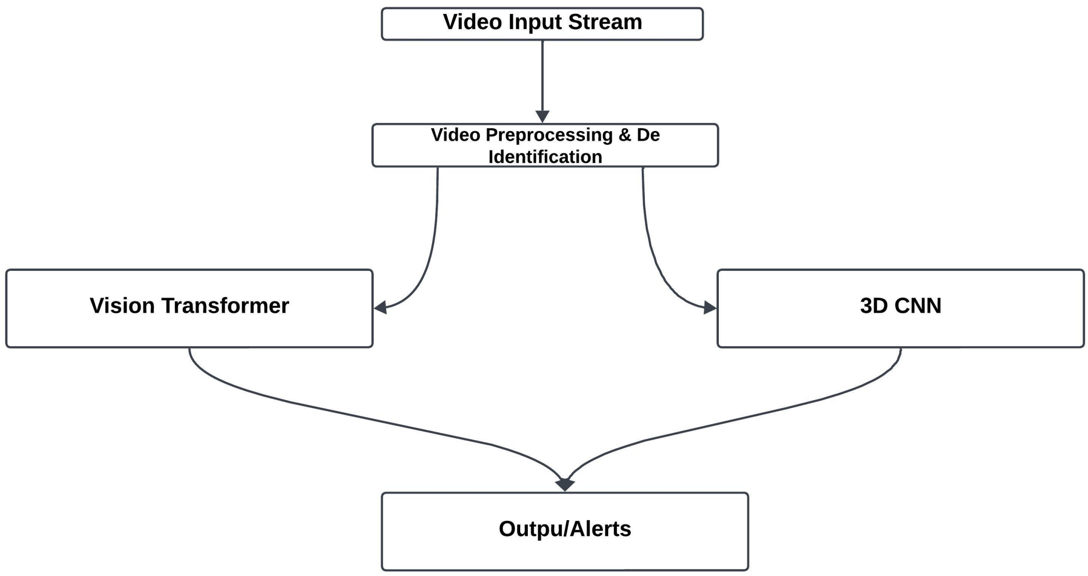

Figure 3.1: Potential Architecture

Description

This diagram highlights a feasible structure of SurgiGuide, which contains Vision Transformer (ViT), 3D Convolutional Neural Network (CNN), and Random Forest models which gain benefits from each one of them.

1. **Video Input Stream:** SurgiGuide receives a continuous stream of video frames from cameras in the operating room.

2. **Video Preprocessing & De-identification:** This stage involves preparing the video data for analysis by:

    - Noise reduction to improve image quality
    - Frame rate adjustment to optimize processing and maintain real-time performance
    - Surgical field isolation to focus on the relevant surgical area

3. **ViT Model:** ViT would be the one responsible for frame by frame and sequence by sequence analysis of the frames in the surgical video. It does a good job of capturing what is spatial information, what is present in each frame as well as temporal relation, the changes of the scene over a period. This is accomplished through the input of both the RGB frames for detailed image characteristics and the optical flow frames for activity patterns. This model developed from a set of surgeries matched with surgical outcomes and complications; aims at identifying the possible risks of developing any adverse event during the current surgery.

4. **3D CNN Model:** Due to their ability to capture temporal sequences, 3D CNNs are appropriate for phase recognition in SurgiGuide. Herein, ViT would offer a general perspective of the scene typical for surgery, whereas the 3D CNN would offer instance-specific details tracing the temporal dynamics of the surgical process.

5. **Output/Alerts:** The results of both models are combined to give real-time feedback to the surgeon.

Bibliography

[1] D. Arad, A. Rosenfeld, and R. Magnezi. Factors contributing to preventing operating room “never events”: a machine learning analysis. *Patient Safety in Surgery*, 17:6, 2023.

[2] T. Igaki, D. Kitaguchi, and H. Matsuzaki. Automatic surgical skill assessment system based on concordance of standardized surgical field development using artificial intelligence. *JAMA Surgery*, 158(8):e231131, 2023.

[3] D. Kiyasseh, R. Ma, and T. F. Haque. A vision transformer for decoding surgeon activity from surgical videos. *Nature Biomedical Engineering*, 7:780–796, 2023.

[4] Yunfan Li, Himanshu Gupta, Prateek Prasanna, IV Ramakrishnan, and Haibin Ling. Surgical phase recognition in laparoscopic cholecystectomy. *Procedia Computer Science*, 239:2006–2012, 2024.

[5] M. Ortenzi, J. Rapoport Ferman, and A. Antolin. A novel high accuracy model for automatic surgical workflow recognition using artificial intelligence in laparoscopic totally extraperitoneal inguinal hernia repair (tep). *Surgical Endoscopy*, 37:8818–8828, 2023.

[6] M. Wagner, B. Müller-Stich, A. Kisilenko, D. Tran, P. Heger, L. Mündermann, D. M. Lubotsky, B. Müller, T. Davitashvili, M. Capek, A. Reinke, C. Reid, T. Yu, A. Vardazaryan, C. I. Nwoye, N. Padoy, X. Liu, E. Lee, C. Disch, and S. Bodenstedt. Comparative validation of machine learning algorithms for surgical workflow and skill analysis with the heichole benchmark. *Medical Image Analysis*, 86:102770, 2023.